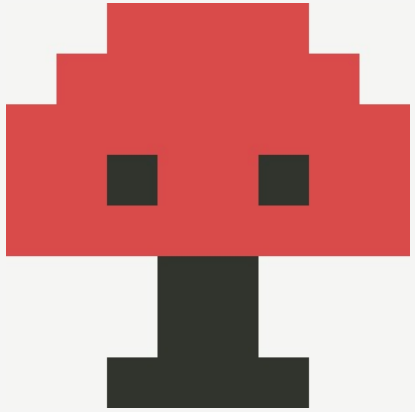
Training AI to Pixelate with Code

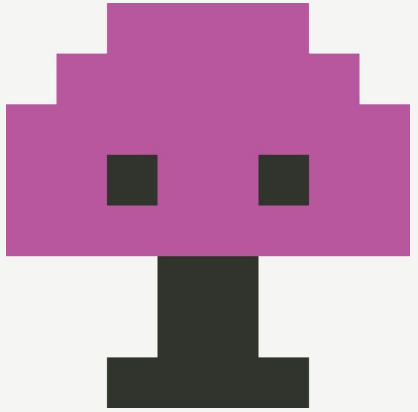
Team 7 · Adam Torres Encarnacion

Make a drawing. Keep the code editable.



Original

→



Change the color, keep the shape

Working reference example, not model output

Problem + parent papers: 5 min · Method + evaluation: 5 min

Three papers, one proposed workflow

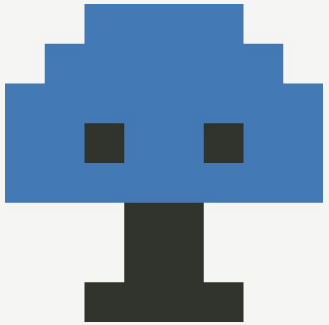
AutomaTikZ ICLR 2024	Trains language models to turn text descriptions into editable scientific figures written in TikZ. Keep code with its rendered image
Text2Reward ICLR 2024	Uses a language model to write reward code from task descriptions for training reinforcement-learning agents. Make technical checks explicit
Diffusion-DPO CVPR 2024	Uses human choices between image pairs to fine-tune a diffusion model for visual appeal and prompt matching. Compare pictures; train on code

These papers motivate the project. They do not prove it works for pixel art.

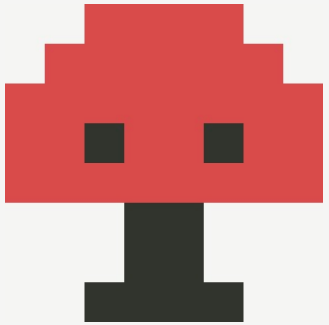
A good-looking picture can still be wrong

Will preference tuning improve pixel art
without breaking the code or simple edits?

Request: a red mushroom



Appealing, but the wrong color



Correct color and readable shape

Hypothesis: people prefer tuned outputs, with validity and editability preserved.

Illustrative reference programs, not a model comparison

Learn from the picture; update the code model

Qwen2.5-Coder-7B-Instruct

01 → 02 → 03 → 04

Two programs

Same prompt

Render + check

Log invalid code

Human choice

Keep both programs

Token-level DPO

LoRA + frozen reference

Use language-model DPO, not a diffusion denoising loss.

Implemented: renderer, checks, and pair export. Training is proposed.

A small dataset with a held-out comparison

120 prompts

100 train / 20 test

5 objects × 6 colors × 4 positions

32 × 32 canvas · fixed scene splits

Base model vs. preference-tuned model

20 test prompts × 3 seeds = 60 comparisons

Valid code + prompt compliance

Human preference + appearance

Recolor without moving other pixels

Count invalid outputs. Keep test references out of training.

Better drawings, with code worth keeping

≥95% valid programs

≥60% preference credit

Proposed targets, not results. Report ties and uncertainty.

Ready: dataset, renderer, checks, and demo

Next: label pairs → train adapter → test

Project code

Dataset

github.com/0xABAN/classes/tree/main/ds-340w